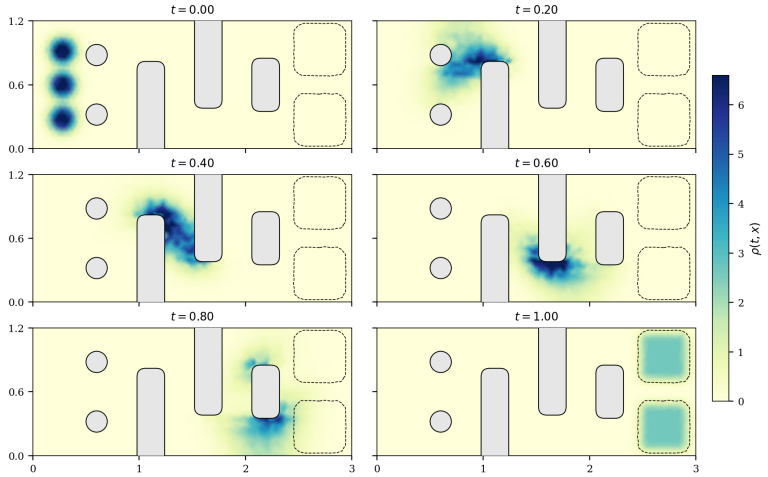
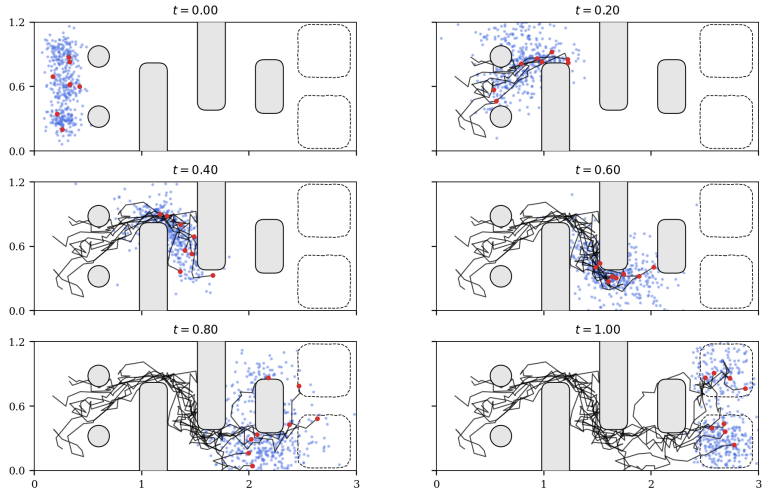
1. Numerical Experiments

Here the experiments are conducted on 2-dimensional domain $[0, 3.0] \times [0, 1.2]$ punctured by 2 circular and 3 rounded rectangular interior holes. The initial and endpoint marginals are, respectively, 3 mixture of Gaussian and split uniform distribution with Gaussian decay at boundary. The nonlocal meanfield is Guassian function $E(x) = \gamma_0 \cdot \exp\left(-\|x\|^2/2\sigma_0^2\right)$ where $\gamma_0 = 0.02$ is strenth and length scale $\sigma_0 = 0.35$. The unstructured mesh is generated by *Gmsh* library with constrained on maximum mesh size $h_{\max}$. Note that there is to noticeable difference between choice of structured or unstructured mesh in execution time or convergence in fixed-point contraction rate. See figure 1 along with convergence 1.



(a) finite-element nodal density $\rho(t, x)$



(b) $N_p = 1600$ independent Brownian particles with 8 highlighted sample paths.

Figure 1: Illustrates the approximated density flow solution $\rho(t, x)$ alongside controlled Euler–Maruyama particle trajectories for baseline parameters $\kappa = 0.12$, $N_t = 50$, and $h_{\max} = 0.06$ ($N = 1388$ nodes). The outer iterate is with damping factor $\theta = 0.65$ converges in $n = 22$ iterations and attained L^1 density residual of $\|\Delta\rho^{22}\|_{L^1} = 1.156 \times 10^{-8}$ ($\|\Delta\rho^{22}\|_{L^\infty} = 3.515 \times 10^{-6}$), a terminal Wasserstein-2 particle discrepancy of $W_2 = 5.365 \times 10^{-2}$, and a total wall-clock execution time 155.47 s.

Firstly, the nonlinear outer iteration is isolated (consider fixed density) and focus purely on linear forward-backward components. Here the artificial construction of frozen fields $b(t, x), r(t, x)$ along with terminal $\varphi(T, x)$ where $x = (x_0, x_1) \in \mathbb{R}^2$ are as follow, :

$$b(t, x) = (\cos x_1, \sin x_0), \;\; r(t, x) = \sin x_0 \cos x_1, \;\; \text{and} \;\; \cos \varphi(T, x) = 1 + \cos x_1 \cos x_0. \tag{1}$$

However, closed form expression exists and so the reference φ_{ref} is computed with finer time-space grid. Doupling termporal reference grid from 640 to 1280 steps (for fixed casrse mesh with $h_{\max} = 0.0787$) alters L^2 weight by 7.741×10^{-1} due to secular spatial interpolation residue. And for $h_{\max} = 0.035$ coupling $N_t = 320$ to 640 changes the field by 1.182×10^{-3}. And changes spatial reference from 0.045 to 0.035 changes the field by 1.502×10^{-2}.

n	$\|\delta\rho^n\|_{L^1}$	$\frac{\|\delta\rho^n\|_{L^1}}{\|\delta\rho^{n-1}\|_{L^1}}$	$\|\rho_0^n - \mu_{\text{init}}\|_{L^1}$	$\|\rho_T^n - \mu_{\text{targ}}\|_{L^1}$	$W_2(\rho_T^n, \mu_{\text{targ}})$	$\max_k \lvert\log Z_k\rvert$
1	1.048×10^{0}	—	6.047×10^{-1}	5.614×10^{-1}	5.285×10^{-1}	2.101×10^{-3}
3	4.160×10^{-1}	0.3500	7.408×10^{-2}	6.877×10^{-2}	1.698×10^{-1}	4.163×10^{-2}
5	5.312×10^{-2}	0.3656	9.075×10^{-3}	8.424×10^{-3}	5.704×10^{-2}	3.031×10^{-2}
10	2.227×10^{-3}	0.3496	4.766×10^{-5}	4.425×10^{-5}	3.885×10^{-3}	2.865×10^{-2}
15	1.169×10^{-5}	0.3500	2.503×10^{-7}	2.324×10^{-7}	2.693×10^{-4}	2.864×10^{-2}
16	4.093×10^{-6}	0.3500	8.762×10^{-8}	8.133×10^{-8}	1.574×10^{-4}	2.864×10^{-2}

Table 1: Fixed-point convergence for baseline 2D maze experiment. Boundary marginal L^1 errors are evaluated on the damped iterate $\rho^n = (1-\theta)\rho^{n-1} + \theta\rho^n_{\text{half}}$. every step.

Thus the reported smaller error to reference is due to artifact. Given two successive refinement levels $m-1$ and m the empirical order of convergence (EOC) is measure using $\text{EOC} = \log\left(\text{err}^m/\text{err}^{m-1}\right) / \log\left(h^m_{\max}/h^{m-1}_{\max}\right)$ and mass-weighted nodal norm (computed over reference node $i \in I_{\Delta x}$ are

$$\text{err}_p(\varphi) = \left(\sum_i m_i |\varphi_{\text{interp}}(0,x_i) - \varphi_{\text{ref}}(0,x_i)|^p\right)^{1/p}, \quad \text{err}_\infty(\varphi) = \max_i |\varphi_{\text{interp}}(0,x_i) - \varphi_{\text{ref}}(0,x_i)|.$$

Additionally the the adjoint property, L^1 and Wasserstein-2 are mornitored for terminal distribution matching. Figure 3 show 1-dim Mckean-Vlasov solution bridge under extreme diffusion from $\kappa = 1$ to $\kappa = 0.0004$ and repulsive interaction $\gamma_0 = 0$ to $\gamma_0 = +0.8$ values. Under $\kappa \to 0$ the entropic interpolation reduce to narrow deterministic Monge-kantorivich optimal transport type that bifurcate sharply. The Péclet number $\text{Pe}_h = \|b\|_\infty y/(2\kappa) \gg 1$ is increased with slower contraction rate. For large $\gamma_0 > 0$, high density is penalized since $\log W = -(E \star \rho)/\kappa$ and forces mass to spread for unit of time rapidly toward every direction and approaching boundary $x = \pm 1$. This results in more frequent Skorokhod reflection, however constraction property still retained. This is possible when Skorkhod reflection is handled well, for instance that every escaping point is detected and correctly map back to feasible region $\mathcal{D}$ while retaining order of spatial accuracy. This this consideration, a substeping is employed. However for attractive force $\gamma_0 < 0$ which makes particle tend toward dense concentrated area. When the ratio $|\gamma_0|/\kappa$ exceeds the critical bifurcation threshold (e.g. $\gamma_0 = -0.35, \kappa = 0.015$) the nonolocal feedback amplifies density perturbations and the normalized map enters a continually oscillatating cycle for all four metrics.

The choice for Δt in semi-lagrangian P1 scheme must takes into account (1) the local temporal splitting error and (2) the spatial interpolation error accumulated. Setting Δt too small relative to $h_{\max}$ in such a way that Courant number $C_{\text{CFL}} = \|b\|_\infty \Delta t/h_{\max} \to 0$ and diffusion characteristic jump size $\sqrt{6\kappa\Delta t} \ll h$, the stochastic next point $x_i + b(x_i)\Delta t + \sqrt{2\kappa\Delta t}\xi_q$ will be trapped inside their host element $K \in \text{T}_h$. And as $T/\Delta t \to \infty$ the repeated convex barycentratric averaging (interpolation) within one same triangle acts as artificial numerical viscosity of magnitude $h^2_{\max}/\Delta t$. This is confirmed in table 2 that under refining time grind on $h_{\max} = 0.0459$ the EOC stagnates at 0.53 and 0.58 and turn negative EOC rates to -0.32 and -0.34.

Conversely, for too large Δt relative to maze size can induce boundary penetration i.e. the characteristic displacement exceeds the "thickness" of boundary from which it penetrates. This also observed also as for non-convex and narrow interior holes, a single-step sign-distance projection (without taken into account the previous jump position) of $\hat{y} = y - 2\chi(y)\nabla\chi(y)\mathbf{1}_{\{\chi(y)<0\}}$ fails to detect penetration if the half step lands entirely across the interior or project the foot onto the wrong face when crosses the medial axis. Fro instance suppose the wall of width 0.26 with $h_{\max} = 0.04$ and $C_{\text{CFL}} = 10$, one stepproduces $\Delta x = C_{\text{CFL}} h_{\max} = 0.40 > 0.26$ completely tunnelling trhough the hole without even triger subroutine check for $\chi < 0$. This phenomena is exacerbated as $t \to T$ by steep lagarithmic gradient $\nabla \log \varphi_T$ which, in this my setting for Gaussian-decayed distribution μ_{targ} for idstnace $d = 0.5$ units from target, the terminal potential gradient pull strongly the particles with magnitude $\|\nabla \log \varphi_T\|$ approximately $d/\sigma_T^2 = 200$ times, where σ_T^2 is length strength parameter. This give exfective potential penalty of 200×0.5 and physical control drift $\alpha^* = 2\kappa\|\nabla \log \varphi_T\| = 48$ unit per sec. However the domain must also be well-constructed and ensure there is path joining the target to initial distribution.
The figure 5 illustrate this idea. For disconnected domain $[-1,0] \cup [1,2]$ and large terminal gradient blow-up. Here μ_{init} concentrated around first half of the domain while μ_{targ} is on the otherside. It is obvious that there is no solution such that $\rho_T = \mu_{\text{targ}}$ is any sense. Here the Skorokhond reflection at $\{-1,0\}$ and $\{1,2\}$ prevent characteristic path from crossing infeasible domain $\mathbb{R} - \mathcal{D} = (0,1)$. And at the same time the terminal steep Gaussian tail distibution $\nabla \log \varphi_T \approx (x_T - x)/\sigma_T^2 \in [355, 591]$. For $t \ll T$ the (backward) heat semigroup $\varphi_t = \mathcal{U}_{t,T}\varphi_T$ exponentially damps non-constant mode on left half domain at the spectral-gap rate $e^{-\pi^2\kappa(T-t)}$ and maintain flat potential norm gradient $\|\nabla \log \varphi_t\|_{L^\infty(\text{left half of } \mathcal{D})} \ll 1$ over at the begining of t. As $t \to T$ the control will try to match the terminal boundary condition and violently leat to frequent reflection.

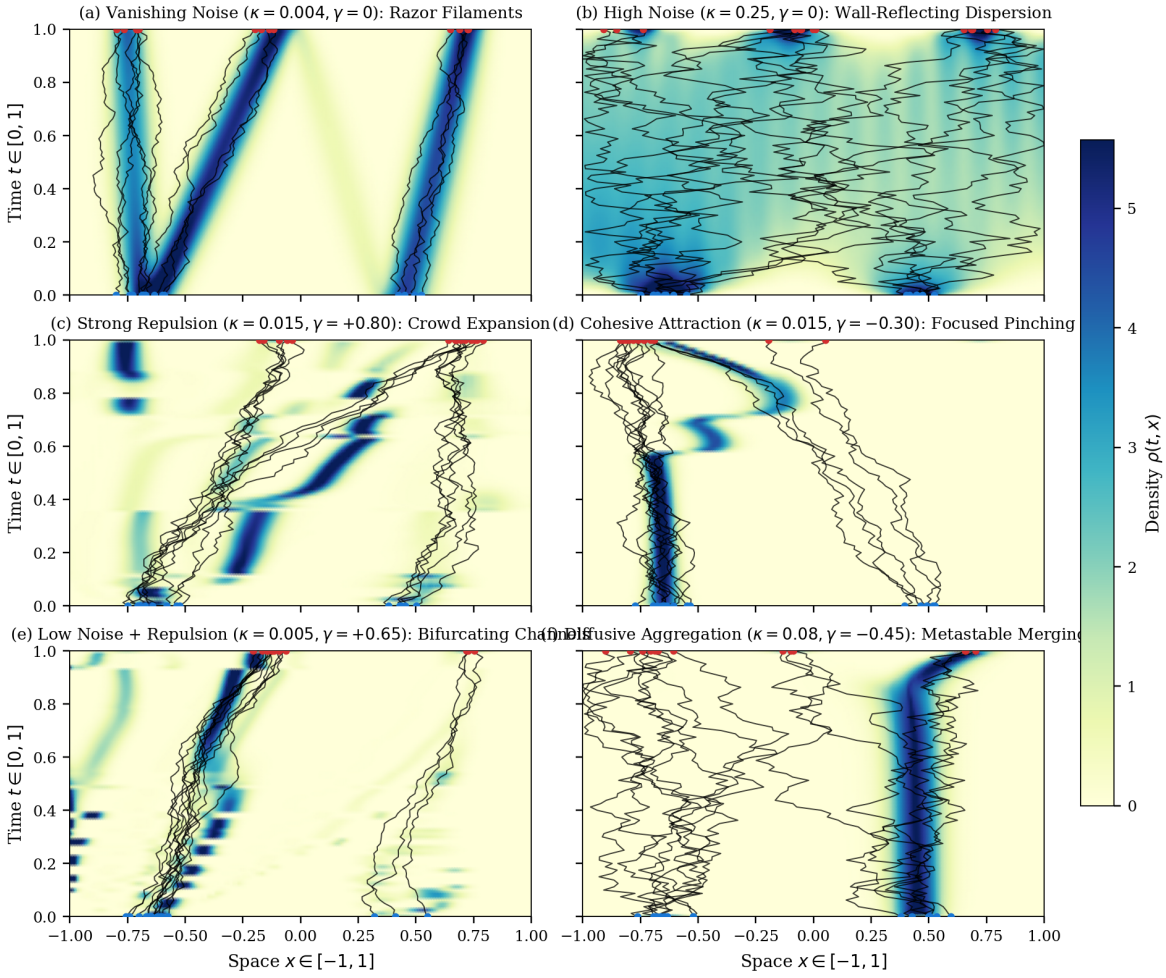


Figure 2: One-dimensional space-time transport plans and particle trajectories under six extreme parameter regimes, illustrating deterministic filamentation ($\kappa = 0.004$), boundary-reflecting dispersion ($\kappa = 0.25$), repulsive crowd expansion ($\gamma = +0.80$), and attractive cohesive pinching ($\gamma = -0.30, -0.45$).

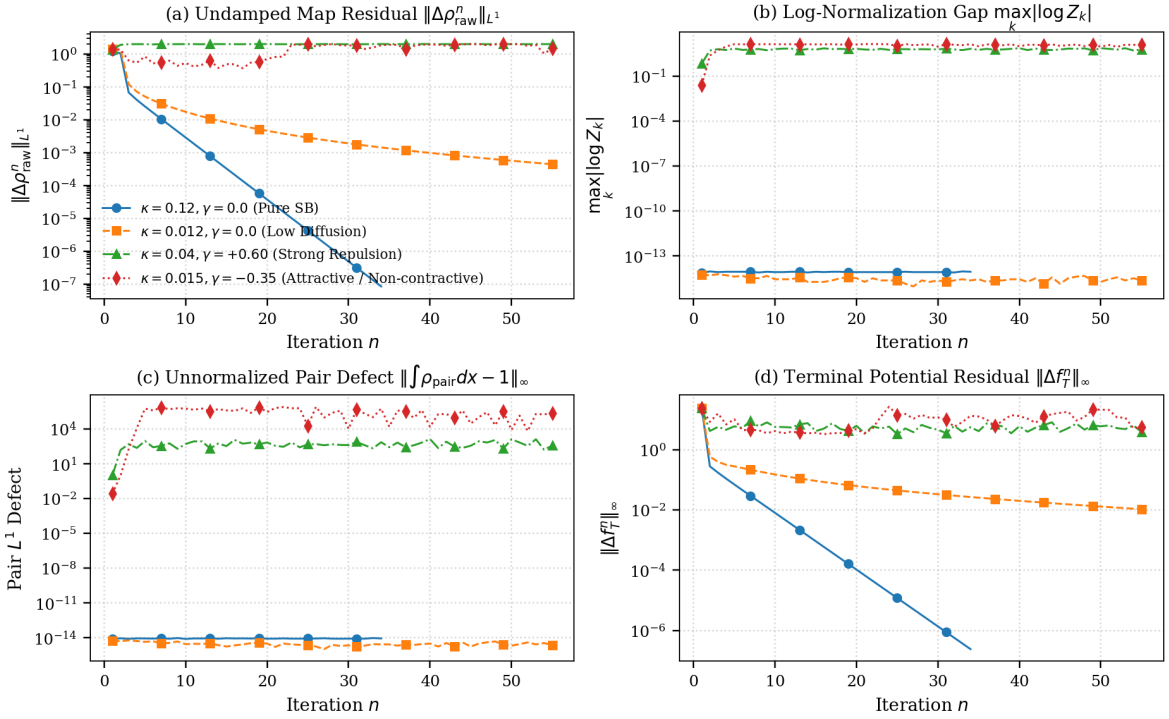


Figure 3: Convergence diagnostics across (κ, γ) regimes: (a) undamped map residual $\|\Delta\rho^n_{\text{raw}}\|_{L^1}$; (b) log-partition gap $\max_k|\log Z_k|$; (c) unnormalized pair-density L^1 defect; (d) terminal potential residual $\|\Delta f^n_T\|_\infty$.

Table 2: Error convergence and computational runtime for the inner linear solve with frozen field.

Δt	$h_{\max}$	$\lvert\mathcal{I}_{\Delta x}\rvert$	$e_\infty(\phi)$	$e_2(\phi)$	$e_2(\nabla\phi)$	EOC	CPU (s)
Temporal Refinement (fixed $h_{\max}=0.0459$)							
0.2000	0.0459	3344	3.492×10^{-1}	1.288×10^{-1}	1.255×10^{0}	—	0.37
0.1000	0.0459	3344	2.073×10^{-1}	7.240×10^{-2}	6.747×10^{-1}	0.83	0.38
0.0500	0.0459	3344	1.321×10^{-1}	4.125×10^{-2}	3.615×10^{-1}	0.81	0.25
0.0250	0.0459	3344	7.355×10^{-2}	2.830×10^{-2}	2.054×10^{-1}	0.54	0.30
0.0125	0.0459	3344	4.480×10^{-2}	1.966×10^{-2}	1.179×10^{-1}	0.53	0.44
0.0083	0.0459	3344	3.468×10^{-2}	1.552×10^{-2}	7.961×10^{-2}	0.58	0.59
0.0063	0.0459	3344	1.945×10^{-2}	1.228×10^{-2}	5.487×10^{-2}	0.81	0.74
0.0042	0.0459	3344	1.385×10^{-2}	6.513×10^{-3}	2.527×10^{-2}	1.56	0.96
0.0028	0.0459	3344	3.266×10^{-3}	2.512×10^{-3}	8.525×10^{-3}	2.35	1.25
Spatial Refinement (fixed $\Delta t=0.0063$)							
0.0063	0.2019	324	1.829×10^{-1}	1.429×10^{-1}	4.237×10^{-1}	—	0.03
0.0063	0.1576	455	1.084×10^{-1}	8.772×10^{-2}	3.019×10^{-1}	1.97	0.05
0.0063	0.1067	678	7.907×10^{-2}	4.170×10^{-2}	1.900×10^{-1}	1.91	0.13
0.0063	0.0787	1388	5.428×10^{-2}	1.258×10^{-2}	7.914×10^{-2}	3.93	0.24
0.0063	0.0408	4567	2.794×10^{-2}	1.548×10^{-2}	6.703×10^{-2}	-0.32	1.04
0.0063	0.0068	142105	4.853×10^{-2}	2.818×10^{-2}	1.405×10^{-1}	-0.34	37.61
Spatiotemporal Refinement							
0.0833	0.2019	324	2.405×10^{-1}	6.436×10^{-2}	2.932×10^{-1}	—	0.02
0.0500	0.1576	455	1.856×10^{-1}	4.168×10^{-2}	2.423×10^{-1}	1.76	0.03
0.0312	0.1067	678	1.413×10^{-1}	2.551×10^{-2}	1.715×10^{-1}	1.26	0.05
0.0182	0.0787	1388	4.195×10^{-2}	1.450×10^{-2}	1.075×10^{-1}	1.85	0.14

Note that the full nonlinear solver incured the discrete adjoint error for forward backward propagations almost to 10^{-15} for all steps. Notably the damped outer density residual $\left\|\rho^n - \rho^{n-1}\right\|_{L^1}$ across different spatial operator splitting Lie vs Strang or choice of quadrature stecils including axial4,axial5,gh9, and monte-carlo MC give very identical convergence curve as illustrated in graph 4.

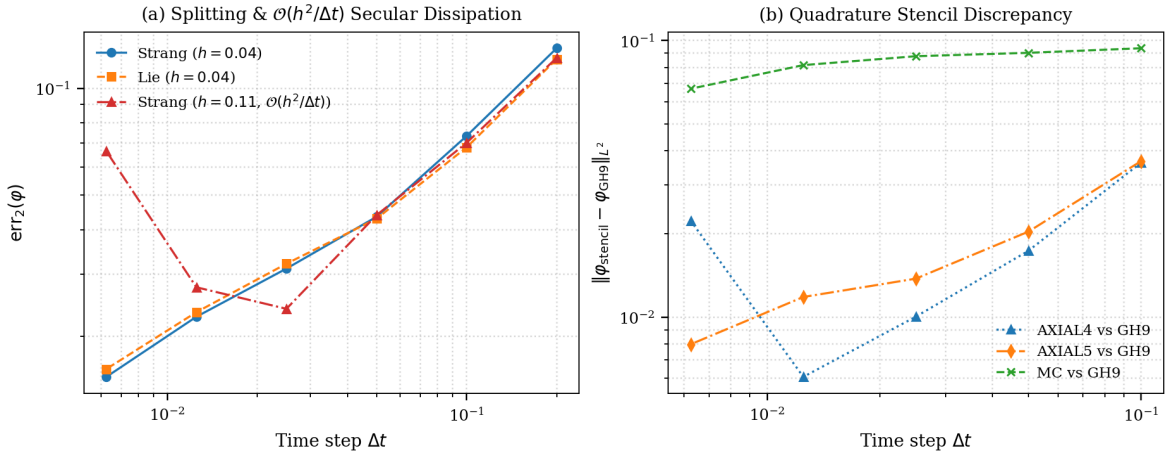


Figure 4: (a) $\mathrm{err}_2(\varphi)$ versus Δt illustrating second-order Strang versus first-order Lie scaling on a fine mesh ($h_{\max} = 0.04$) alongside the $\mathcal{O}(h^2/\Delta t)$ secular dissipation upturn on a coarse mesh ($h_{\max} = 0.11$); (b) L^2 discrepancy of `axial4`, `axial5`, and 32-sample antithetic Monte Carlo stencils relative to the tensor-product benchmark `GH9`.

Furthermore, adopting a log-domain implementation significantly eliminates floating-point overflow, i.e. setting $f = \log \varphi, g = \log \hat{\varphi}$. This is particularly important for low diffusion level (e.g. $\kappa < 0.02$) which when evaluate raw density (which create order of $\exp\left(-\|x-y\|^2/(4\kappa\Delta t)\right) \sim e^{-100}$) causes underflow and division-by-zero in Sinkhorn scaling map. Using log-domain also helps with small negative oscillations near sharp boundary due to dispersion induced by P1 dispersion, which can cause Birkhoff-Bushell contraction to fail. In other word the normalize map $\rho = \exp(\log \rho - \log Z) > 0$ is ensured by construction positive strictly.

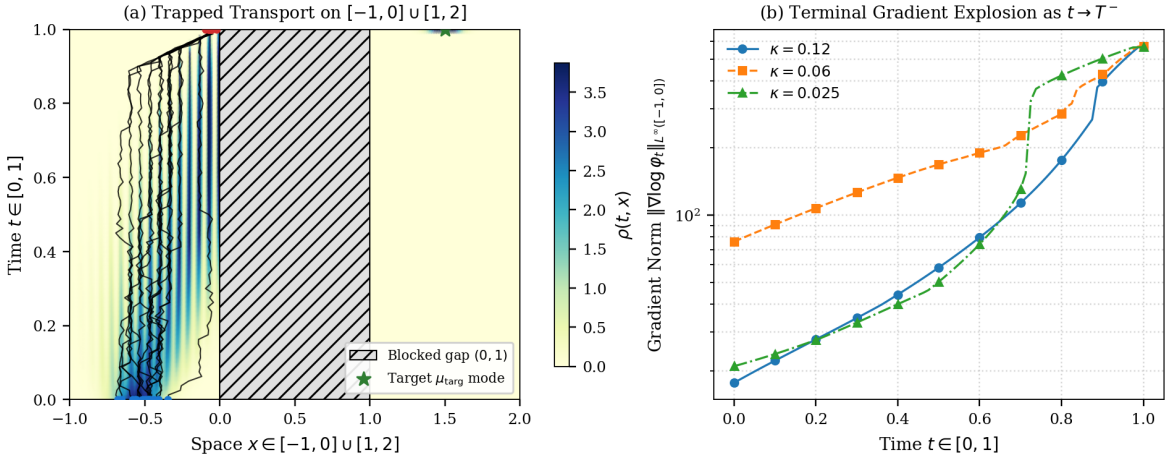


Figure 5: Topological obstruction and terminal gradient blow-up on the disconnected domain $\Omega = [-1, 0] \cup [1, 2]$: (a) particles initialized in μ_{init} on $[-1, 0]$ are blocked by the forbidden gap $(0, 1)$ and accumulate in a boundary layer at $x = 0^-$ near $t \to T^-$; (b) temporal evolution of $\|\nabla \log \varphi_t\|_{L^\infty([-1,0])}$ showing exponential blow-up as $t \to T^-$.

2. Conclusion